

Mirage Mini-Split IR Protocol Documentation

This document was derived from captures taken from the original remote control using ESPHome's Mirage protocol decoder. The protocol appears to be a 14-byte state-based packet.

Packet Structure

Byte 0: Header (0x56)
Byte 1: Temperature (temp + 0x5C)
Byte 2: Unknown / Reserved
Byte 3: Quiet flag
Byte 4: Mode + Fan encoding
Byte 5: Sweep, Display, Power flags
Byte 6: Health, Sleep, Heat flags
Byte 7: Clean flag
Byte 8: Turbo flag
Bytes 9-13: Unused / Reserved

Temperature Encoding

Formula: Byte1 = Temperature + 0x5C

Examples:

20°C = 0x70

21°C = 0x71

Mode Encoding

Heat = 0x10

Cool = 0x20

Dry = 0x30

Auto = 0x40

Fan = 0x50

Fan Speed Offsets

Auto = +0

High = +1

Low = +2

Medium = +3

Examples

Cool + Auto = 0x20

Cool + High = 0x21

Cool + Low = 0x22

Cool + Medium = 0x23

Flags

Byte 3

0x01 = Quiet Mode

Byte 5

0x01 = H-Sweep

0x02 = V-Sweep

0x03 = H+V Sweep

0x04 = Display Toggle

0xC0 = Power Off

Byte 6

0x02 = Health

0x08 = Sleep

0x0A = Health + Sleep

0x40 = Heat Flag

Byte 7

0x40 = Clean

Byte 8

0x80 = Turbo

Captured Reference Commands

Baseline Cool 20°C Auto Fan:

56:70:00:00:20:00:00:00:00:00:00:00

Heat:

56:73:00:00:11:00:40:00:00:00:00:00

Power Off:

56:70:00:00:20:C0:00:00:00:00:00:00

Turbo:

56:70:00:00:20:00:00:00:80:00:00:00

Clean:

56:70:00:00:20:00:00:40:00:00:00:00